Enabling AI safety with NeMo Guardrails and Evaluate LLMs with EvalHub

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Home

Introduction

Course Title: Enabling AI safety with NeMo Guardrails and Evaluate LLMs with EvalHub

Description: This course covers the hands-on lab on the NVIDIA NeMo Guardrails and Garak (Generative AI Red-teaming & Assessment Kit) Evaluation Framework. This hands-on lab focuses on how to use NeMo Guardrails and Garak Evaluation Framework features in the Red Hat OpenShift AI (RHOAI) environment.

Duration: …

Objectives

On completing this course, you should be able to:

- Deploy and test NeMo Guardrails by using only built-in detectors and no LLM calls
- Deploy NeMo Guardrails with an LLM for advanced guardrail capabilities including self-check rails
- Test model security against known attacks using tools like Nvidia Garak

Prerequisites

This course assumes that you have the following prior experience:

- A basic understanding of machine learning principles and workflows is required
- A basic understanding of Generative AI and Large Language Models (LLMs) is required
- Basic experience with Git is required
- Experience with containers and Red Hat OpenShift is recommended, or completion of the Red Hat OpenShift Developer II: Building and Deploying Cloud-native Applications (DO288) course

Contributors

The PTL team acknowledges the valuable contributions of the following Red Hat associates:

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Lab Environment

Instructions to Launch Your Lab on the Red Hat Demo Platform (RHDP)

1. **Log in** to the RHDP portal
2. **Search** for the catalog entry: **Red Hat OpenShift Container Platform Cluster (Multi-Cloud)**.
3. **Click** on the catalog entry in the search results.
4. On the catalog page, **click** the **Order** button.
5. **Fill out** the required details in the order form.
   a. Cloud Provider - cnv
   b. OpenShift Version 4.20
   c. Cluster size - multinode
   d. OpenShift Worker Count - 2
   e. OpenShift Worker CPU - 16
   f. OpenShift Worker Memory - 64Gi
6. **Review the warning** at the bottom of the form and **check the box** labeled:
   “I confirm that I understand the above warnings.”
7. **Click** the **Order** button to place your lab order.

Important Notes:

- This lab may take approximately **90 minutes** to become ready.
- You will receive an **email with access details** once your lab environment is ready.
- You can also **retrieve lab access** directly from the RHDP portal.

How to Access Your Lab via the RHDP Portal:

1. On the RHDP portal, **click** on the **Services** option in the left-hand menu.
2. **Select your lab** from the listings on the right-hand side of the page to view access details.

RHDP Portal Links

- RedHat associates: https://demo.redhat.com/
- RedHat partners: https://partner.demo.redhat.com/

NeMo Guardrails

In this chapter, focus is on the following objectives:

- Deploy and test NeMo Guardrails by using only built-in detectors and no LLM calls
- Deploy NeMo Guardrails with an LLM for advanced guardrail capabilities including self-check rails

What are NeMo Guardrails?

NeMo Guardrails Configuration for Lemonade Stand (No Guardrails)

NeMo Guardrails Configuration - Lemonade Stand with Regex Detection

NeMo Guardrails Configuration - Lemonade Stand with Regex Detection and PII

NeMo Guardrails Configuration - Lemonade Stand with LLM-as-Judge + Regex Detection + PII

NeMo Guardrails Configuration - Lemonade Stand with Multi-Model Guardrails + Regex Detection + PII + HAP

NeMo Guardrails Configuration - Lemonade Stand with Multi-Model Guardrails + Regex Detection + PII + HAP + Custom Actions

NeMo Guardrails Configuration - Lemonade Stand with Multi-Model Guardrails + Regex Detection + PII + HAP + Custom Actions + Language Detection

This page demonstrates `@asciidoctor/tabs` in Antora: use tabs for multiple languages, or to separate exercise instructions from solutions.

Language tabs

Java

```
public class Hello {
  public static void main(String[] args) {
    System.out.println("Hello");
  }
}
```

Python

```
def main():
    print("Hello")

if __name__ == "__main__":
    main()
```

Exercise and solution

Exercise

Try writing a function that returns the sum of two integers. Use your language’s usual entry point if you need a quick test.

Solution

```
function add(a, b) {
  return a + b;
}
console.log(add(2, 3));
```

Industry-specific self-check guardrail examples

This page demonstrates collapsible example blocks in Antora: use them for optional **Hints**, **Deep dives**, or other detail that should not block the main lesson flow.

No extra Antora extension is required; `[%collapsible]` is built into Asciidoctor.

Hint (collapsed by default)

▼ *Hint*

Try the exercise first; open this only if you are stuck.

- Check that your environment variables are set.
- Re-read the previous section on configuration.

Deep dive (collapsed by default)

▼ *Deep dive: how the pipeline stages relate*

The build runs **validate**, then **compile**, then **package**. Skipping a stage can leave artifacts out of date; that is why CI always runs the full chain.

```
./gradlew clean build
```

Starts expanded

▼ *Optional context*

This block uses `[%collapsible%open]` so it is expanded on first load. Use sparingly when the extra context is still optional but you want it visible by default.

GARAK (Generative AI Red-teaming & Assessment Kit)

In this chapter, focus is on the testing model security against known attacks using tools like Nvidia Garak.

Evaluating AI Systems

Deploy EvalHub with the TrustyAI Operator